SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

CO1-ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Annexure A – Application Based Questions

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name. Identify
email addresses.
Extract mobile numbers.
Detect technical skills (Python, Java, SQL, Machine Learning, NLP). Extract
years of experience.
Generate a structured summary of the candidate's profile.
Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

CODE:

```
import re

resumes = [
    """Name: Arun Kumar
    Email: arun@gmail.com
    Mobile: 9876543210
    Skills: Python, Java, SQL
    Experience: 3 years""",

    """Name: Priya Sharma
    Email: priya@gmail.com
    Mobile: 9123456780
    Skills: SQL, NLP
    Experience: 1 year""",

    """Name: Rahul Raj
    Email: rahul@gmail.com
    Mobile: 9988776655
    Skills: Python, Machine Learning
    Experience: 2 years"""
]

for resume in resumes:
    name = re.search(r"Name:\s*(.+)", resume)
    email = re.search(r"[\w.-]+@[\w.-]+\.\w+", resume)
    mobile = re.search(r"\b[6-9]\d{9}\b", resume)
    exp = re.search(r"Experience:\s*(\d+)\s*years?", resume, re.I)

    skills = re.findall(
        r"Python|Java|SQL|Machine Learning|NLP",
        resume, re.I
    )

    years = int(exp.group(1)) if exp else 0
```

```
python_skill = bool(re.search(r"\bPython\b", resume, re.I))

print("\n--- Candidate Profile ---")
print("Name:", name.group(1).strip() if name else "Not found")
print("Email:", email.group() if email else "Not found")
print("Mobile:", mobile.group() if mobile else "Not found")
print("Skills:", ", ".join(skills) if skills else "None")
print("Experience:", years, "years")

if years >= 2 and python_skill:
    print("Status: Eligible")
else:
    print("Status: Not Eligible")
```

OUTPUT:

```
PS C:\Users\nancy> & C:/Users/nancy/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/nancy/npl_1.py

--- Candidate Profile ---
Name: Arun Kumar
Email: arun@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL
Experience: 3 years
Status: Eligible

--- Candidate Profile ---
Name: Priya Sharma
Email: priya@gmail.com
Mobile: 9123456780
Skills: SQL, NLP
Experience: 1 years
Status: Not Eligible

--- Candidate Profile ---
Name: Rahul Raj
Email: rahul@gmail.com
Mobile: 9988776655
Skills: Python, Machine Learning
Experience: 2 years
Status: Eligible
PS C:\Users\nancy>
```

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.
Perform prefix-based searches.
Perform suffix-based searches.
Search using partial keywords.
Perform case-insensitive searches.
Display all matching product names.
Generate a report showing the total number of matching products for each search.

CODE:

```
import re

products = [
    "Python Book",
    "Python Laptop",
    "Wireless Mouse",
    "Gaming Laptop",
    "Python USB Drive",
    "Laptop Stand"
]

def search(keyword, search_type):
    if search_type == "exact":
        pattern = r"^" + re.escape(keyword) + r"$"
    elif search_type == "prefix":
        pattern = r"^" + re.escape(keyword)
    elif search_type == "suffix":
        pattern = re.escape(keyword) + r"$"
    elif search_type == "partial":
        pattern = re.escape(keyword)
    else:
        print("Invalid search type")
        return

    matches = [
        p for p in products
        if re.search(pattern, p, re.I)
    ]

    print("\nSearch Type:", search_type)
    print("Keyword:", keyword)
    print("Matching Products:")

    for p in matches:
        print(p)

    print("Total Matches:", len(matches))
```

search("Python Laptop", "exact")
search("Python", "prefix")
search("Laptop", "suffix")
search("Laptop", "partial")
search("python", "partial")

OUTPUT:

```
KeyboardInterrupt
PS C:\Users\nancy> & C:/Users/nancy/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/nancy/npl_1.py

  Search Type: exact
  Keyword: Python Laptop
  Matching Products:
  Python Laptop
  Total Matches: 1

  Search Type: prefix
  Keyword: Python
  Matching Products:
  Python Book
  Python Laptop
  Python USB Drive
  Total Matches: 3

  Search Type: suffix
  Keyword: Laptop
  Matching Products:
  Python Laptop
  Gaming Laptop
  Total Matches: 2

  Search Type: partial
  Keyword: Laptop
  Matching Products:
  Python Laptop
  Gaming Laptop
  Laptop Stand
  Total Matches: 3

  Search Type: partial
  Keyword: python
  Matching Products:
```

You have Docker installed on
install the recommended ext

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

Validate the student register number.
Validate the institutional email address.
Validate the course code.
Validate the semester information. Validate
the student's mobile number.
Display appropriate validation messages for each field.
Generate a final registration status report indicating whether the registration is successful.

CODE:

```
import re

reg = input("Enter Register Number: ")
email = input("Enter Institutional Email: ")
course = input("Enter Course Code: ")
semester = input("Enter Semester: ")
mobile = input("Enter Mobile Number: ")

v1 = bool(re.fullmatch(r"\d{2}[A-Z]{3}\d{3}", reg))
v2 = bool(re.fullmatch(r"[\w.-]+@university\.edu", email, re.I))
v3 = bool(re.fullmatch(r"[A-Z]{2,4}\d{3}", course))
v4 = semester.isdigit() and 1 <= int(semester) <= 8
v5 = bool(re.fullmatch(r"[6-9]\d{9}", mobile))

print("\n--- Validation Report ---")
print("Register Number:", "Valid" if v1 else "Invalid")
print("Institutional Email:", "Valid" if v2 else "Invalid")
print("Course Code:", "Valid" if v3 else "Invalid")
print("Semester:", "Valid" if v4 else "Invalid")
print("Mobile Number:", "Valid" if v5 else "Invalid")

if all([v1, v2, v3, v4, v5]):
    print("\nRegistration Successful")
else:
    print("\nRegistration Failed")
    print("Please correct the invalid fields.")
```

OUTPUT:

```
PS C:\Users\nancy>
PS C:\Users\nancy> & C:/Users/nancy/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/nancy/npl_1.py
Enter Register Number: 192324117
Enter Institutional Email: nancy.sse@saveetha.com
Enter Course Code: dsa0311
Enter Semester: 7
Enter Mobile Number: 9676667211

--- Validation Report ---
Register Number: Invalid
Institutional Email: Invalid
Course Code: Invalid
Semester: Valid
Mobile Number: Valid

Registration Failed
Please correct the invalid fields.
PS C:\Users\nancy>
```

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____